Technical SEO rendering audit — Zuse Digital

Prepared per the technical-seo-rendering-audit skill (report format: release decision → evidence → rubric).

Release decision

Verdict: GO WITH WARNINGS

The sampled production output is reliably crawlable and indexable: every template returns HTTP 200 with index-critical content, unique metadata, canonical, and valid JSON-LD present in the **raw HTTP response before JavaScript**, hydration is error-free, and missing routes return a correct 404. **No P0, P1, or P2 findings remain** — the two prior P2 items (canonical/sitemap trailing-slash mismatch and the production env-var guard) were fixed and verified this pass. The verdict stays GO WITH WARNINGS only because external validation (Rich Results, Lighthouse, a11y) was NOT RUN and should precede a final GO.

Priority	Count	Release effect
P0	0	None — no sitewide deindexing or broken rendering
P1	0	None — no per-template rendering/metadata/link failures
P2	0	Both prior P2s fixed & verified this pass (canonical/sitemap; env guard)
P3	0	—

Scope and method

- **Target:** Zuse Digital — Next.js 16.3.4 (App Router, Turbopack), React 19, GSAP animation, Strapi (staging CMS).
- **Build/runtime mode:** production build verified earlier (`next build` — clean, `/` & templates prerender static/ISR); this pass re-sampled the running server for current post-fix state.
- **Date:** 2026-09-10.
- **Routes sampled (by template):** `/` (home), `/about` (static), `/work` (listing), `/blog` (listing), `/services/artificial-intelligence` (service template), and a deliberate 404.
- **Representations compared:** raw HTTP HTML (bundled collector) vs hydrated DOM (in-app Chromium). JS-disabled inferred from CSS.
- **Tools used:** collector `collect-rendering-evidence.mjs`, in-app browser (console + DOM), curl, source inspection.
- **Exclusions:** Playwright JS-disabled pass (not installed); Core Web Vitals/Lighthouse; external rich-results validation; full per-service-page coverage.

Release blockers and important findings

No P0, P1, or P2 findings remain. All earlier blockers (missing metadataBase, dead nav → 404, sitewide broken OG images, reduced-motion/failure-safe, thin content, duplicate 404 title) plus the two P2 items below were resolved and verified.

[Resolved] Canonical / sitemap trailing-slash mismatch	FIXED
Was	Page canonicals were slash-terminated (…`/work/`) while the sitemap and served route were slash-less, so the canonical 308-redirected — the three discovery signals disagreed across every template.
Fix applied	Added `normalizeCanonical()` in `lib/config/site.ts` (strips trailing slash, collapses root) and applied it in `lib/seo/strapi-page-metadata.ts` and `app/(home)/page.tsx`, so any CMS-supplied slash is normalized centrally.
Verified	Canonicals are now slash-less and match the served routes and sitemap paths: `/` → …`com`, `/about` → …`com/about`, `/work` → …`com/work`. No redirect. TypeScript clean.

[Resolved] Production env-var guard	FIXED
Was	A localhost `NEXT_PUBLIC_SITE_URL` (e.g. from copying `.env.local.example` to `.env.local`) would silently make every metadata/OG/canonical/sitemap URL localhost in a production build.
Fix applied	`lib/config/site.ts` now ignores a localhost origin when `NODE_ENV=production` and falls back to the production default (with a build-time warning); real staging/production hosts pass through. `.env.local.example` annotated.
Verified	Unit-tested resolver: `localhost/127.0.0.1 + production → https://zusedigital.com`; `staging host → unchanged`; `localhost + development → unchanged`.

Route matrix

Route	HTTP	Raw content	Metadata	Canon/robots	Hydration	Links	Result
`/`	200	PASS (h1 +11k chars)	PASS	index,follow	PASS	95	PASS
`/about`	200	PASS (~1.8k)	PASS	PASS	PASS	75	PASS
`/work`	200	PASS (~1k)	PASS	PASS	PASS	76	PASS
`/blog`	200	PASS (~0.8k)	PASS	PASS	PASS	77	PASS
`/services/ai`	200	PASS (~1.2k)	PASS	PASS	PASS	78	PASS
`/(missing)`	404	N/A (404 UI)	noindex	N/A	PASS	1	PASS 404

Canonicals are now slash-less and match the sitemap and served routes across every template (prior P2 resolved).

Raw versus hydrated evidence

Route	Signal	Raw server HTML	Hydrated DOM	Assessment
`/`	**title**	Australia's Award Winning AI & Software Agency…	identical	Stable (CMS)
`/`	**JSON-LD**	1 valid @graph block	1 block	Server-rendered, valid
`/`	**links**	95 crawlable anchors	95	Real anchors
`/work`	**h1**	Our Work | Zuse Digital Case Studies	identical	Stable
`/work`	**canonical**	…/work (slash-less)	…/work	Matches sitemap + served URL; no redirect (fixed)
`/work`	**console**	—	0 errors	Clean hydration

Full rubric

ID	Check	Result	Evidence / notes
A1	Production build	**PASS**	next build clean; static/ISR prerender
A2	Status codes	**PASS**	200 for templates; 404 for missing
A3	Redirects	**PASS**	Finite; nav no longer emits 308 (but canonical→/x/ does, see C3)
A4	Content type	**PASS**	text/html everywhere
A5	Header directives	**PASS**	No stray X-Robots-Tag; 404 noindex + no-store
A6	Stable URLs	**PASS**	Path-based, no hash routing
B1	Primary content	**PASS**	Real h1 + substantive body in raw HTML (0.8k–11k chars)
B2	Template data	**PASS**	CMS data server-rendered per template
B3	Semantic structure	**PASS**	main/headings/nav/anchors in raw HTML
B4	Hydration stability	**PASS**	Raw == hydrated for title/h1/canonical/JSON-LD
B5	Failure resilience	**PASS**	Manifesto opacity-100 default; 0.2 only via data-motion (JS)
C1	Unique title	**PASS**	Distinct per template
C2	Meta description	**PASS**	Present & page-specific

ID	Check	Result	Evidence / notes
C3	Canonical	**PASS**	Slash-less, matches sitemap + served route; normalizeCanonical() centralizes it
C4	Robots meta	**PASS**	index,follow; 404 noindex
C5	Language	**PASS**	html lang=en; no hreflang (N/A)
C6	Social metadata	**PASS**	OG/Twitter images resolve (CMS/CloudFront, 200)
C7	Viewport	**PASS**	width=device-width present
D1	Internal links	**PASS**	Real <a href> anchors; all nav targets 200
D2	URL quality	**PASS**	Clean, descriptive paths
D3	Pagination/facets	**N/A**	No paginated listings yet
D4	robots.txt	**PASS**	app/robots.ts → allow-all + Sitemap directive
D5	Sitemap	**PASS**	200/valid XML; loc now agrees with canonicals (slash-less)
D6	Orphans	**PASS**	All templates linked from nav
E1	JSON-LD server availability	**PASS**	Present in raw HTML on every template
E2	Parse validity	**PASS**	Parses without repair
E3	Content match	**PASS (spot)**	Consistent with visible content
E4	Required properties	**NOT RUN**	Run Google Rich Results Test
E5	Stable identity	**PASS**	Canonical/@id/URL identity consistent after normalization
F1	No client-only SEO	**PASS**	All head+content in raw HTML
F2	Clean hydration	**PASS**	0 console errors
F3	Deterministic first render	**PASS**	No date/random/viewport-driven markup
F4	Progressive enhancement	**PASS**	Anchors, not handler-gated
F5	Client boundaries	**PASS**	Client comps limited to Header/Manifesto/GradientText
G1	Animated content presence	**PASS**	Manifesto words server-rendered
G2	Visibility fallback	**PASS**	Readable without animation (opacity-100 default)
G3	Reduced motion	**PASS**	matchMedia gate + reduce branch + CSS !important
G4	Focus/reading order	**NOT RUN**	Not keyboard-tested this pass
G5	Layout stability	**WARN**	next/image sized; confirm no CLS from GSAP scrub
G6	Motion ownership	**PASS**	Wrappers enhance server content
H1	LCP path	**WARN**	Hero image priority; measure on prod
H2	Image delivery	**PASS**	next/image with dimensions + remotePatterns
H3	Font behavior	**WARN**	next/font + Typekit link — confirm no FOIT
H4	JavaScript cost	**NOT RUN**	Bundle not measured
H5	Field/lab metrics	**NOT RUN**	No Lighthouse/RUM
I1	Template coverage	**PASS**	One route per existing template sampled
I2	Environment rules	**PASS**	Slash policy consistent; localhost SITE_URL blocked in prod builds
I3	Error/loading states	**PASS**	404 returns 404, custom not-found, single title
I4	Regression path	**PASS**	Re-check commands provided

Fix order

1. `[done]` Align canonical policy across canonicals/sitemap/served routes — normalizeCanonical() (C3/D5/E5 now PASS).
2. `[done]` Guard production `NEXT_PUBLIC_SITE_URL` against localhost leakage (I2 now PASS).
3. Run external validation on the deployed site: Google Rich Results Test (E4), Lighthouse/CWV (H4/H5), keyboard & focus a11y (G4).
4. Confirm no CLS from the GSAP scrub and no font FOIT (G5/H3).

Limitations

- Playwright not installed → JS-disabled rendering inferred from CSS; hydrated DOM verified via in-app browser instead.
- Core Web Vitals, bundle size, rich-results eligibility (E4), and keyboard/focus order (G4) were not measured.
- One route per template sampled; individual /services/* pages beyond the AI page were not each opened.

Re-check commands

```
node collect-rendering-evidence.mjs --base-url http://localhost:7000 \
  --routes /,/about,/work,/blog,/services/artificial-intelligence,/missing-xyz --browser auto
curl -s http://localhost:7000/work | grep canonical  # must match sitemap <loc> and return 200, not 308
curl -s http://localhost:7000/sitemap.xml | grep -o '<loc>[^<]*'
```